

APIx — Real-time Airfare Price Index

Official Technical Approach & Architectural Specification

Target Beneficiaries:	Ministry of Statistics & Programme Implementation (MoSPI), RBI, Aviation Economists
GitHub Repository:	https://github.com/ybsolanki/apix-airfare-index
Live Platform URL:	https://ybsolanki.github.io/apix-airfare-index/
Core Stack:	Python FastAPI, SQLite, React, Vite, Tailwind CSS, Recharts, Pandas, NumPy

1. Executive Summary & Problem Context

Traditional Consumer Price Index (CPI) calculations rely on static monthly manual surveys. In modern aviation, dynamic pricing algorithms adjust airfares multiple times daily based on demand, booking windows, fuel prices, and seat inventory. This creates significant lag and misalignment between official CPI figures and actual consumer airfare expenditure.

APIx Solution: APIx establishes an automated, high-frequency Airfare Price Index engine that ingests, cleans, and normalizes domestic airfare observations across 24 representative Indian flight corridors, computing a real-time weighted index relative to base period Jan 2026 = 100.0.

2. Technical System Architecture

- **Ingestion Layer (scrapers/):** Abstract BaseScraper adapter framework extracting published economy airfares across direct airline portals (IndiGo, Air India, Akasa Air, SpiceJet) and OTAs (MakeMyTrip, Yatra, EaseMyTrip, Cleartrip, Ixigo).
- **Data Quality Engine (processors/):** Outlier detection (IQR filters for fares < ₹1,000 or > ₹60,000), cabin class normalization (15kg economy standard), and multi-OTA deduplication.
- **Index Engine (index/calculator.py):** Mathematical Laspeyres index relative to base period t0 = 100.0, weighted by route seat capacity.
- **REST API & Dashboard (api/ & apix-frontend/):** Python FastAPI backend serving OpenAPI JSON endpoints and a responsive React analytics dashboard.

3. Mathematical Index Methodology

Route Price Relative:	$I_{r,t} = (P_{r,t} / P_{r,0}) \times 100$
National Aggregate Index:	$I_{total,t} = \sum (w_r \times I_{r,t}) \text{ [where } \sum w_r = 1.0 \text{]}$

Advance Purchase Horizons (Yield Management Standardizing):

Observations are tracked across 6 advance booking horizons (T+1, T+3, T+7, T+15, T+30, T+45) to isolate pure price inflation from dynamic last-minute surge pricing.

4. Domestic Indian Route Dataset Coverage (24 Corridors)

Route Corridor	Category	Distance	Base Fare (t0)	Route Weight (wr)
Delhi → Mumbai (DEL-BOM)	Metro	1,148 km	■5,400	0.095 (9.5%)
Mumbai → Delhi (BOM-DEL)	Metro	1,148 km	■5,450	0.095 (9.5%)
Delhi → Bengaluru (DEL-BLR)	Metro	1,740 km	■6,200	0.080 (8.0%)
Bengaluru → Delhi (BLR-DEL)	Metro	1,740 km	■6,250	0.080 (8.0%)
Mumbai → Bengaluru (BOM-BLR)	Metro	842 km	■4,300	0.065 (6.5%)
Bengaluru → Mumbai (BLR-BOM)	Metro	842 km	■4,350	0.065 (6.5%)
Delhi → Kolkata (DEL-CCU)	Metro	1,305 km	■5,600	0.055 (5.5%)
Mumbai → Ahmedabad (BOM-AMD)	Tier-2	441 km	■3,200	0.040 (4.0%)
Delhi → Hyderabad (DEL-HYD)	Metro	1,253 km	■5,100	0.050 (5.0%)
Delhi → Goa / Kochi / Pune	Tier-2	1,100+ km	■5,300 - ■7,400	0.095 (9.5%)

5. Automated Test Suite & Verification Results

- **Pytest Suite (apix-backend/tests/test_api.py):** 100% Success (7/7 unit tests passed).
- **Advance Purchase Pricing Rule:** Verified T+1 average fare (■11,090) > T+45 average fare (■5,906).
- **Vite Build & GitHub Pages:** Production web application built and deployed live at <https://ybsolanki.github.io/apix-airfare-index/>.